

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases. Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC)

Feature	HDD	SSD
Speed	Slower because it uses a spinning disk.	Much faster because it uses memory chips.
Durability	Less durable. It can be damaged if dropped.	More durable. It can handle shocks and drops better.
Storage	Gives more storage at a lower price.	Costs more but gives better performance.
Noise	Makes a little noise while working.	Works silently because it has no moving parts.
Typical Use	Used in desktop PCs and for storing large files.	Used in laptops, gaming PCs, and fast computers.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

➤ **CPU (Central Processing Unit):**

The CPU is the brain of the computer. It controls the game and processes all the instructions. It handles player actions, enemy movements, and game logic. A faster CPU helps the game run smoothly without slowing down.

➤ **RAM (Random Access Memory)**

RAM stores the game data that the computer needs while you are playing. It helps the game load faster and keeps it running smoothly. If the PC has more RAM, the game can run better and switch between tasks more easily.

➤ **GPU (Graphics Processing Unit)**

The GPU is responsible for showing the game's graphics on the screen. It creates the characters, maps, buildings, and special effects. A powerful GPU gives smoother gameplay, higher FPS, and better picture quality, making games like PUBG or Free Fire look more realistic.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

➤ **RAM:**

I would upgrade the RAM to at least **16 GB**. More RAM helps the game run smoothly and reduces lag while playing.

➤ **Graphics Card (GPU):**

I would upgrade the GPU because it shows the game's graphics. A better GPU gives better picture quality, smoother gameplay, and higher FPS.

➤ **SSD (Solid State Drive):**

I would install an SSD if my PC has an HDD. An SSD makes the game load faster and also helps Windows start quickly.

➤ **Processor (CPU):**

I would upgrade the CPU if it is old or slow. A faster CPU helps the game run better and handles all the game tasks smoothly.

➤ **Power Supply (SMPS):**

If I install a new GPU or CPU, I may also upgrade the SMPS. It provides enough power for all the computer parts to work properly.